2AMS11 Survival Analysis for Data Scientists
Group Assignment: Survival Analysis Project

Week 3 to Week 8

Q1, 2026

1 Introduction

In this group assignment (3-4 students), you will develop, implement, and evaluate a survival model for a realistic application. The project is divided into three parts. Each part has a deadline for submission (see Canvas). After Parts I and II, you will receive feedback. You incorporate this feedback in the next submission. Your grade will depend both on the quality of your new work and on how well you incorporate the feedback into subsequent submissions (see grading section at the end).

You may use large language models (LLMs) such as ChatGPT, Copilot, Claude, Gemini, or similar tools. All use of LLMs must be documented explicitly (see Section 7) in each submission.

Students should work in groups and for each submission contribution statements should be included.

2 Learning Objectives

By completing this project, you should be able to:

- Formulate an appropriate survival model for a practical problem.
- Identify relevant covariates and censoring mechanisms.
- Justify modeling assumptions.
- Design and implement a simulation study.
- Evaluate the consequences of model misspecification.
- Assess estimator performance using bias, variance, and mean squared error (MSE).
- Communicate results clearly and critically.

3 Part I: Model Specification

Task

Choose a realistic situation in which a survival model is appropriate.

Examples include:

- Time until hospital readmission.
- Time until machine failure.
- Time until unemployment ends.

You are free to choose an application.

Required Components

1. **Practical Context**

   Describe:

   - The event of interest.
   - The target population.
   - Why survival analysis is appropriate.

2. **Survival Model**

   Specify a parametric survival model (e.g. exponential, Weibull, log-normal, or log-logistic).

   Explain why the chosen model is appropriate for the proposed application.

3. **Covariates**

   Include at least three covariates.

   For each covariate:

   - Define it clearly.
   - Explain why it is expected to affect survival.
   - State the expected direction of its effect.
   - If relevant, explain the relationships among the covariates

4. **Censoring**

   Describe the censoring mechanism.

   Discuss:

   - Which type of censoring occurs.
   - Why censoring arises in practice.
   - Whether censoring is independent of the event time.

   Provide a clear justification. If independence may not hold, discuss why.

5. **Data-Generating Mechanism**

   Provide a mathematical description of how data will be generated.

   Specify:

   - Parameter values.
   - Covariate distributions.
   - Sample size.
   - Censoring mechanism.

   No simulation results are required at this stage.

6. **Research Question**

   State at least one research question that will be investigated through simulation.

   Examples:

   - How accurately is a treatment effect estimated?
   - How does increasing censoring affect estimator performance?
   - How sensitive are results to model misspecification?

Deliverable

Report should include AI and contribution statement. The length of the report is maximum 1 page, excluding AI and contribution statements.

4 Part II: Simulation Study Design

Task

Design the simulation study that will be used to evaluate your model. The design should be sufficiently detailed that another group could implement the study based solely on your description.

Required Components

1. **Simulation Algorithm**

   Provide a step-by-step description of the simulation procedure.

   For example:

   (a) Generate covariates.
   (b) Generate event times.
   (c) Generate censoring times.
   (d) Construct the observed dataset.
   (e) Fit the proposed model.
   (f) Store quantities of interest.
   (g) Repeat for many simulation runs.

   A flowchart is encouraged.

2. **Software Implementation**

   Provide pseudocode or actual code.

   The code should include:

   - Data generation.
   - Application of censoring.
   - Model fitting.
   - Storage of simulation results.

3. **Performance Measures**

   Specify the quantities that will be evaluated.

   At minimum include:

   - Bias.
   - Variance (or empirical standard error).
   - Mean squared error (MSE).

4. **Alternative (Incorrect) Analysis**

   Propose at least one intentionally incorrect procedure to compare with the correct analysis.

   Possible examples include:

- Omitting an important covariate.
- Ignoring right censoring.
- Fitting an exponential model when data are generated from a Weibull model.
- Assuming independent censoring when censoring is informative.
- Dichotomizing a continuous covariate.

Explain why the procedure is incorrect and what consequences you expect.

5. **Hypotheses**

   State your expectations before conducting the simulation study.

   Examples:

   - Bias will increase when censoring is ignored.
   - MSE will increase under model misspecification.
   - Omitting an important covariate will bias treatment-effect estimates.

Deliverable

Report should include AI and contribution statement. The length of the report is maximum 2 pages, including feedback paragraph, excluding AI and contribution statements.

5 Part III: Simulation Study and Conclusions

Task

Conduct the simulation study proposed in Part 2 and interpret the results.

Required Components

1. **Results**

   Present results using tables, figures, and numerical summaries.

   Clearly distinguish between:

   - The correctly specified analysis.
   - The intentionally incorrect analysis.

2. **Evaluation**

   Discuss the observed:

   - Bias.
   - Variability.
   - MSE.
   - Practical implications.

3. **Comparison with Expectations**

   Compare your findings with the hypotheses formulated in Part 2.

   Discuss:

   - Which expectations were confirmed.

- Which expectations were not confirmed.
- Possible explanations.

4. **Conclusions**

   Provide practical recommendations for applied researchers.

   Discuss:

   - Which assumptions were most influential.
   - Which mistakes should be avoided.
   - The robustness of the analysis to assumption violations.

5. **Limitations**

   Discuss limitations of your simulation study.

   Examples include:

   - Limited sample sizes.
   - Specific parameter choices.
   - Restriction to a particular model family.

Deliverables - UPDATED

Please submit final report and code. For the report the requirements are as follows:
The length of the report on part III depends on how many research questions you have

- 1 research question: 1 page
- 2-3 research questions 1,5 pages
- More than 3 research questions: 2 pages

Also include Part I and Part II reports in your final report and start at a new page for report III.
In addition to the contribution statement for Part III, also provide an overall statement for the four submissions.
As before the length of final report includes feedback paragraph on part II, but excludes reports Part I and II, AI, and contribution statements.

6 Feedback and Assessment

After Parts I and II, feedback will be provided.
For Parts II and III:

- 50% of the points will be based on the quality of the newly completed task.
- 50% of the points will be based on how well feedback from the previous stage has been incorporated.

Each submission after feedback should include a section entitled:

Response to Feedback

In this section, summarize:

- The feedback received.
- How the feedback was addressed.
- Any suggestions that were not adopted and the reasons why.

7 Use of Artificial Intelligence Tools

The use of LLMs is permitted. However, full transparency is required.

Each submission must contain a statement on LLM use. For every use of an LLM, report:

- Date of use.
- Name of the tool.
- Exact prompt(s) used.
- Summary of the output received.
- How the output was used.
- Whether modifications were made.

Students remain fully responsible for the correctness and integrity of all submitted work. Failure to disclose AI use may be considered academic misconduct.

8 Contribution Statement

Each report must end with a contribution statement.

For each group member, describe contributions to:

- Problem formulation.
- Statistical design.
- Programming.
- Analysis.
- Writing and editing.

Example:

> Student A developed the practical application and simulation design.
>
> Student B implemented the simulation code and generated figures.
>
> Student C conducted the statistical analyses and wrote the conclusions.

All group members must confirm that the contribution statement is accurate.

9 Grading Rubric

Part I (5 points)

- Practical motivation and clarity: 1
- Model specification: 1
- Covariates and censoring discussion: 2
- Data-generating mechanism and justification: 1

Part II (10 points)

- Feedback: 5
- Simulation design: 2
- Quality of code/pseudocode: 2
- Choice of performance measures: 1

Part III (10 points)

- Feedback: 5
- Code: 1
- Correct implementation: 1
- Presentation of results: 1
- Interpretation and discussion: 1
- Conclusions and recommendations: 1